Focus Detection Feasibility Gameplan

Nov 7, 2026 · @Mahek

Goal: one person, about 4 hours, Colab Pro. Answer one question: can 8-channel EEG tell attentive from distracted on BBBD? Cheapest test first; stop the moment the answer is clear.

Rules

- Metric: AUC from model scores
- Two splits: **calibrated** (same person, train on one video, test on another) and **new person** (3 participant-grouped folds, not 5)
- Go: calibrated AUC ⩾ 0.70 · Kill: < 0.60

Hour 1: data (Colab)

- Download BBBD Experiment 2 only; add Experiment 3 later only if more data is needed
- Keep our 8 channels, average reference, 1 to 40 Hz, 60 Hz notch
- 2 s windows, attentive = 0 / distracted = 1; save as one .npz per participant, plus a 200 Hz copy

Hour 1.5: Riemannian check (Tang recipe, 20 min)

- 4 bands, covariances, tangent space, logistic regression; runs on CPU in minutes
- **Gate 1:** calibrated ⩾ 0.70 go · 0.60 to 0.70 continue cautiously · < 0.60 rerun with all 64 channels (passes = electrode layout problem; fails = kill, pivot)

Hour 2 to 2.5: CBraMod frozen + probe (GPU)

- Pretrained weights from braindecode, frozen; features per window; logistic regression
- 4 s windows only (more context for the pretrained model)
- **Gate 2:** within 0.05 of Riemannian or better, try fine-tuning; otherwise stop here, Riemannian wins

Hour 2.5 to 3.5: CBraMod fine-tune (only if Gate 2 passes)

- One fold only, early stopping, about 10 epochs on Colab Pro GPU

- Keep it only if ≥ 0.03 above the probe

Hour 3.5 to 4: decide + headset smoke test

- One results table: Riemannian vs. CBraMod probe vs. fine-tuned, both splits
- Headset (30 min, optional): stream 2 min via BrainFlow, check eyes-closed alpha, run the winning model on a 6-min self-calibration (3 min focused, 3 min counting back by 7s)
- Headset too noisy: demo by replaying BBBD through the same pipeline

Cut for time

- Tang reproduction, Bodonhelyi benchmark, cross-dataset test, 5-fold runs, webcam fusion: only after a Go, in later hours

Where it can fail

- Riemannian near 0.5: label or event-timing bug, check before blaming the idea
- 8 channels fail, 64 pass: move electrodes
- Good calibrated, poor new person: product needs per-user calibration (expected)
- CBraMod setup eats time: drop it, the Riemannian result alone answers feasibility